

Summary

- Research Engineer and AI Privacy Specialist with 7+ years of experience applying cutting-edge ML models and LLMs to solve complex security and data problems.
- Working at the intersection of privacy, machine learning, and software architecture.
- Specialized in developing out-of-model defenses, custom RAG systems, and empirical evaluation frameworks for Generative AI applications to build resilient, high-throughput AI infrastructure.

Work Experience

Google

Bangalore, India

AI Privacy Engineer

Jun. 2024 - Present

- Translated complex privacy regulations into technical specifications, designing automated policy enforcement systems and out-of-model defenses for scalable AI deployments.
- Led privacy impact assessments and threat modeling for Generative AI initiatives, proactively engineering mitigations that balanced robust data protection with optimal product functionality.
- Collaborated cross-functionally to integrate privacy-by-design principles throughout the ML product lifecycle, embedding data controls and automated governance into system architectures.

Google DeepMind & Google Research India

Bangalore, India

Pre Doctoral Researcher

Jul. 2022 - Jun. 2024

- **LLM Fine-Tuning & Prompting** - Engineered bulk inference pipelines to generate factual intermediate representations for text-style transfer using large language models (LLMs). Employed prompting, supervised fine-tuning, and LoRA to optimize stylistic accuracy and reduce hallucinations.
- **Multimodal Conversational Interfaces** - Developed a conversational interface over data visualizations using integrated vision-language and large language models. Prioritized accessibility-focused design and privacy-preserving data handling for diverse user needs.

Google through Optimum Infosystems

Bangalore, India

Data Commons Associate

Aug. 2021 - Jun. 2022

- Developed robust data processing workflows for ingesting large-scale statistical datasets (including census data) into the Data Commons platform.
- Designed novel schema for diverse public statistical datasets, significantly expanding data coverage for Google's open knowledge graph project.

Centre for Networked Intelligence, Indian Institute of Science

Bangalore, India

Technical Associate

Dec. 2019 - Jul. 2021

- Led the architecture and development of COVID-19 response projects, including serosurvey optimization platforms and agent-based models as a service.
- Contributed to high-impact epidemiological research published in AAMAS 2021, Journal of IISc, and Sankhya B.

IBM Netherlands

Amsterdam, the Netherlands

Extreme Blue Intern

May. 2018 - Aug. 2018

- Led the development of an AI and data-driven prototype for supply chain analysis at a top Food & Beverage company.
- Received the IBM Benelux Excellence Award and placed as Runner-up for the project with the best innovation impact at the IBM Extreme Blue European Expo.

Selected Publications

- A Talekar, **Shriram, S**, N Vaidhiyan, G Aggarwal, J Chen, S Venkatramanan, L Wang, A Adiga, A Sadilek, A Tendulkar, M Marathe, R Sundaresan, and M Tambe. [Cohorting to Isolate Asymptomatic Spreaders: An Agent-Based Simulation Study on the Mumbai Suburban Railway](#). *Proceedings of the 20th International Conference on Autonomous Agents and MultiAgent Systems*, pages 1680–1682, 2021.
- S Agrawal, S Bhandari, A Bhattacharjee, A Deo, N M Dixit, P Harsha, S Juneja, P Kesarwani, A K Swamy, P Patil, et al. City-scale agent-based simulators for the study of non-pharmaceutical interventions in the context of the covid-19 epidemic. *Journal of the Indian Institute of Science*, pages 1–39, 2020.
- **Sharad, S***, P B Sivakumar, and V Anantha Narayanan. [The smart bus for a smart city—A real-time implementation](#). In *2016 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)*, pages 1–6. IEEE, 2016.
- P Parag, P Patil, **S Shriram**, R Sundaresan, H Tyagi, and N K Vaidhiyan. [Antidote for CoVID-19 pandemic induced surge in online learning](#). In *Proceedings of the ACM SIGCOMM Online Networking Education Discussion Workshop*, 2020.
- S Athreya, G R Babu, A Iyer, M M B S, N Rathod, **Shriram, S**, R Sundaresan, N Vaidhiyan, and S Yasodharan. [COVID-19: Optimal Design of Serosurveys for Disease Burden Estimation](#). *Sankhya B*, pages 1–23, 2020.
- **Sharad, S***, P B Sivakumar, and V Ananthanarayanan. [An automated system to mitigate loss of life at unmanned level crossings](#). *Procedia computer science*, 92:404–409, 2016

Highlights

- Recognized for cross-functional collaboration and architectural leadership with multiple Google internal awards (Peer and Spot bonuses).
- Received the *IBM Benelux Excellence Award* and placed as Runner-up for *Best Innovation Impact* at the IBM Extreme Blue European Expo (2018).

Education

Delft University of Technology (TU Delft)

Delft, the Netherlands

M.S. (by Research) in Computer Science

Sep. 2017 - Aug. 2019

- Courses: Software Architecture, Pattern Recognition, Information Retrieval, Web Data Management, Web Science and Engineering.
- Thesis: "[A Human-Machine Approach to Preserve Privacy in Image Analysis Crowdsourcing Tasks](#)".
- Advisor: Prof. Alessandro Bozzon
- Teaching and Lab Assistant for graduate and undergraduate CS course.

Amrita Vishwa Vidyapeetham (Amrita University)

Coimbatore, India

B.Tech. in Computer Science and Engineering

Aug. 2012 - May. 2016

- Undergraduate Student Researcher in Mobile and Wireless Networks lab.

Skills

AI Security & Privacy	Out-of-Model Defenses, Prompt Injection Mitigation, Contextual Integrity, Threat Modeling
Programming Languages	Python, C++, JavaScript, Java
ML/NLP Frameworks	TensorFlow, Hugging Face, PyTorch, NumPy, LoRA
Data & Infrastructure	Pandas, Apache Beam, SQL, R, Django, Flask